

CHAPTER

4

Electrical Unit Configuration

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4.1 Electrical Unit Configuration

4.1.1 Electrical Unit Configuration

■ Layout Drawing of Electrical Unit

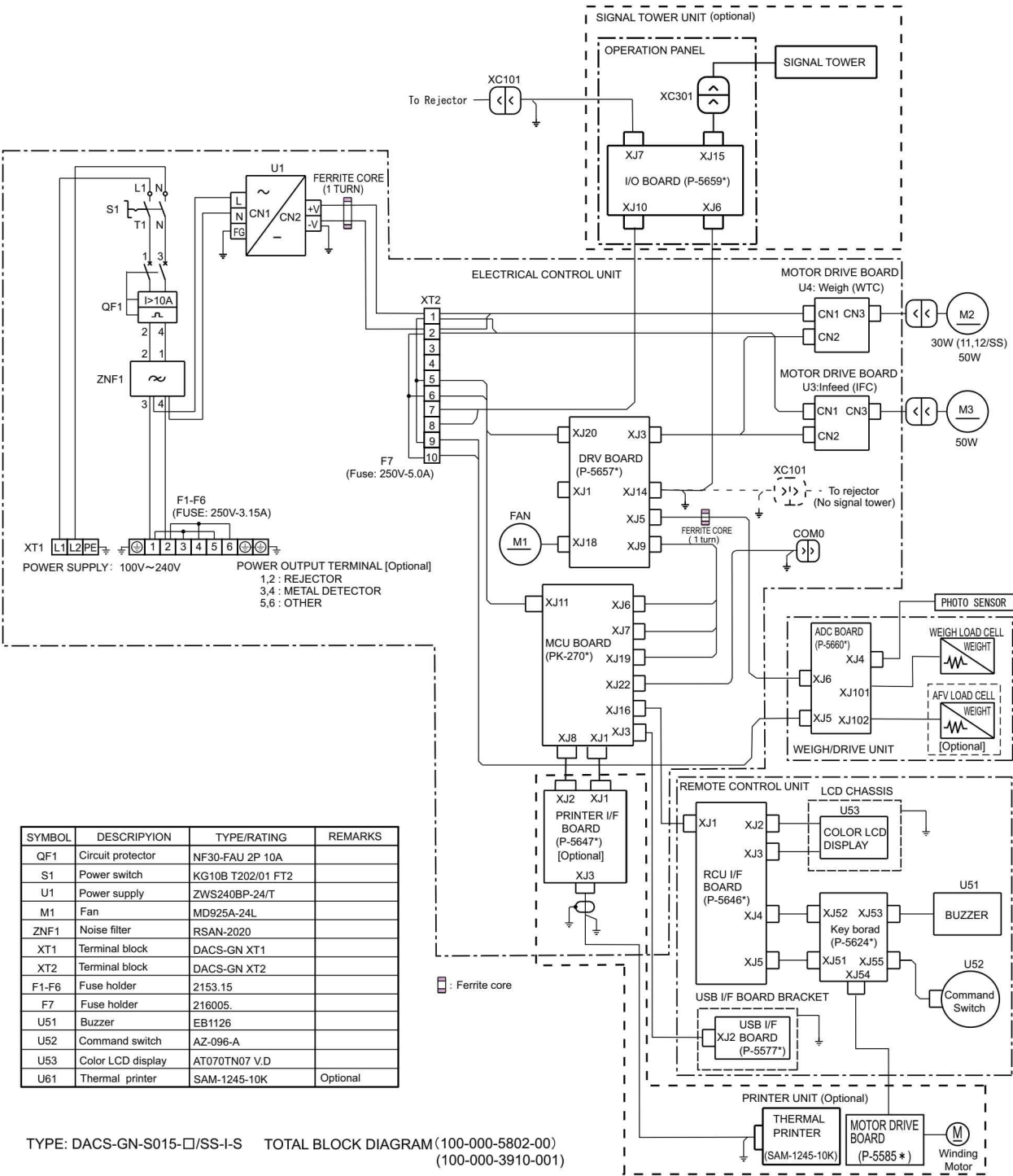


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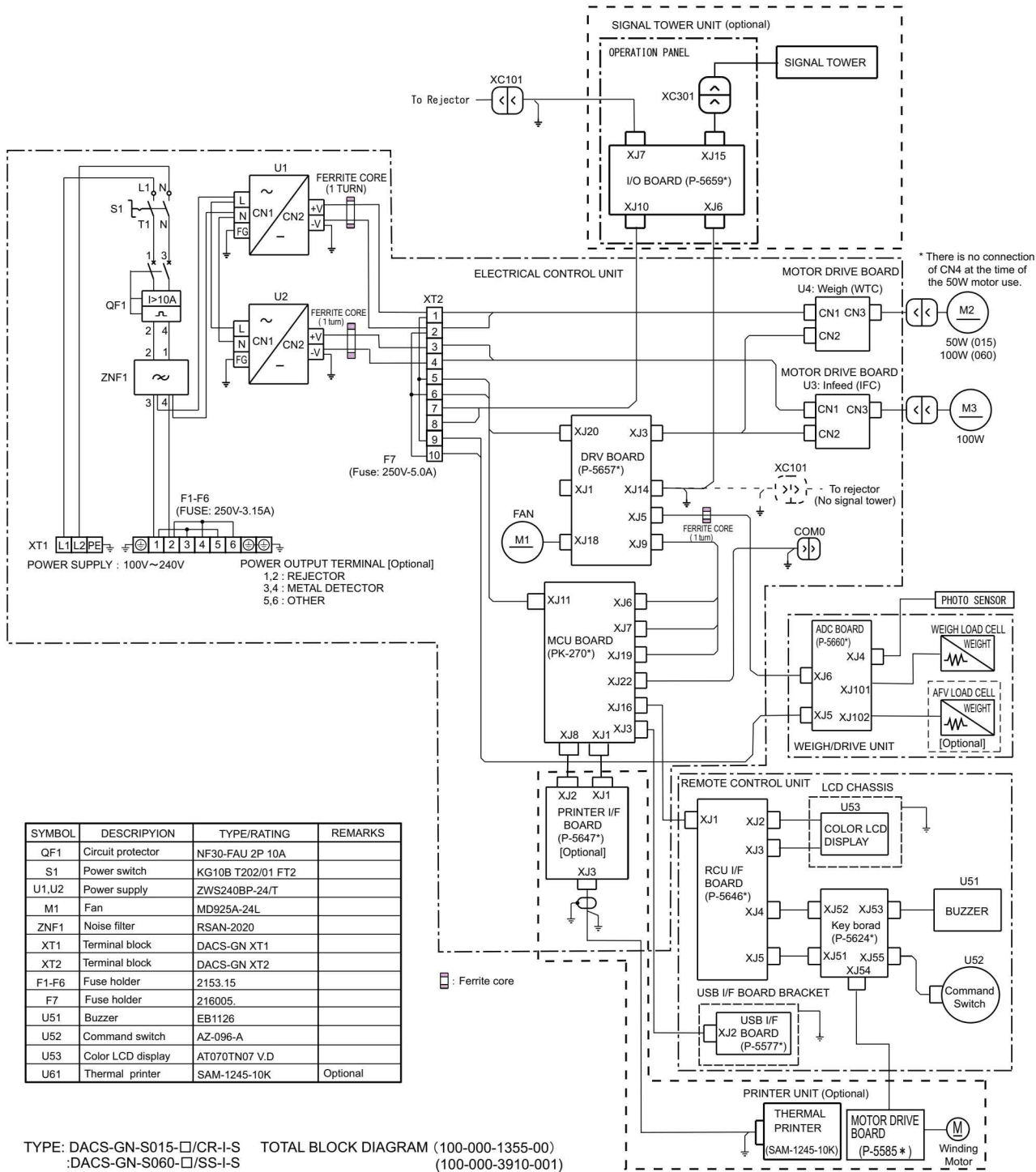
4 Electrical Unit Configuration

4.1.2 Total Diagram

■ DACS-GN-S015-XX/SS-I-S



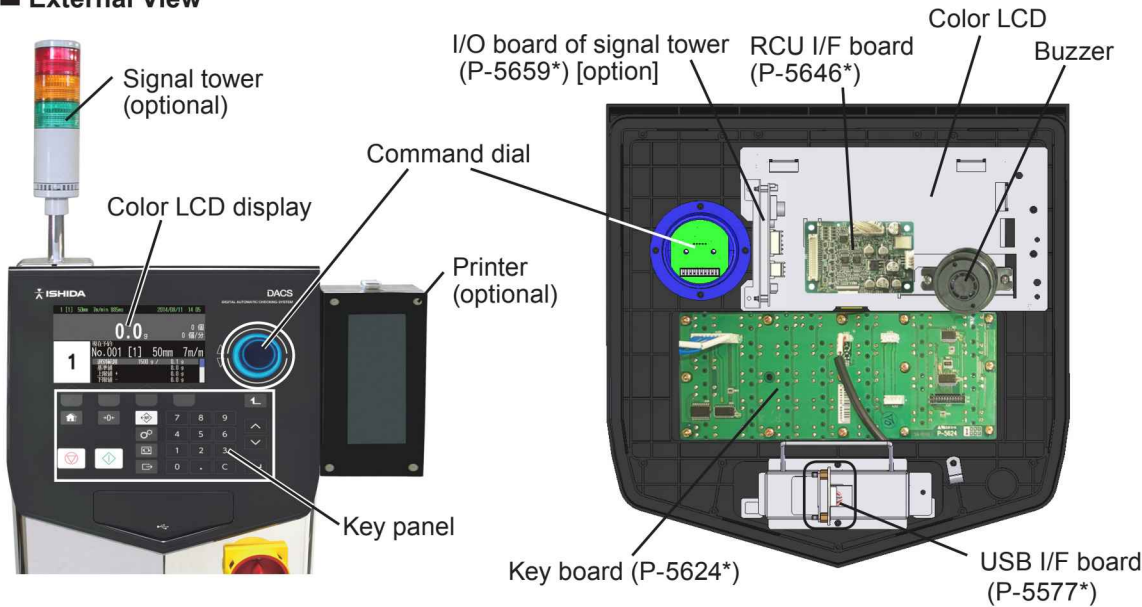
■ DACS-GN-S060-XX/SS-I-S



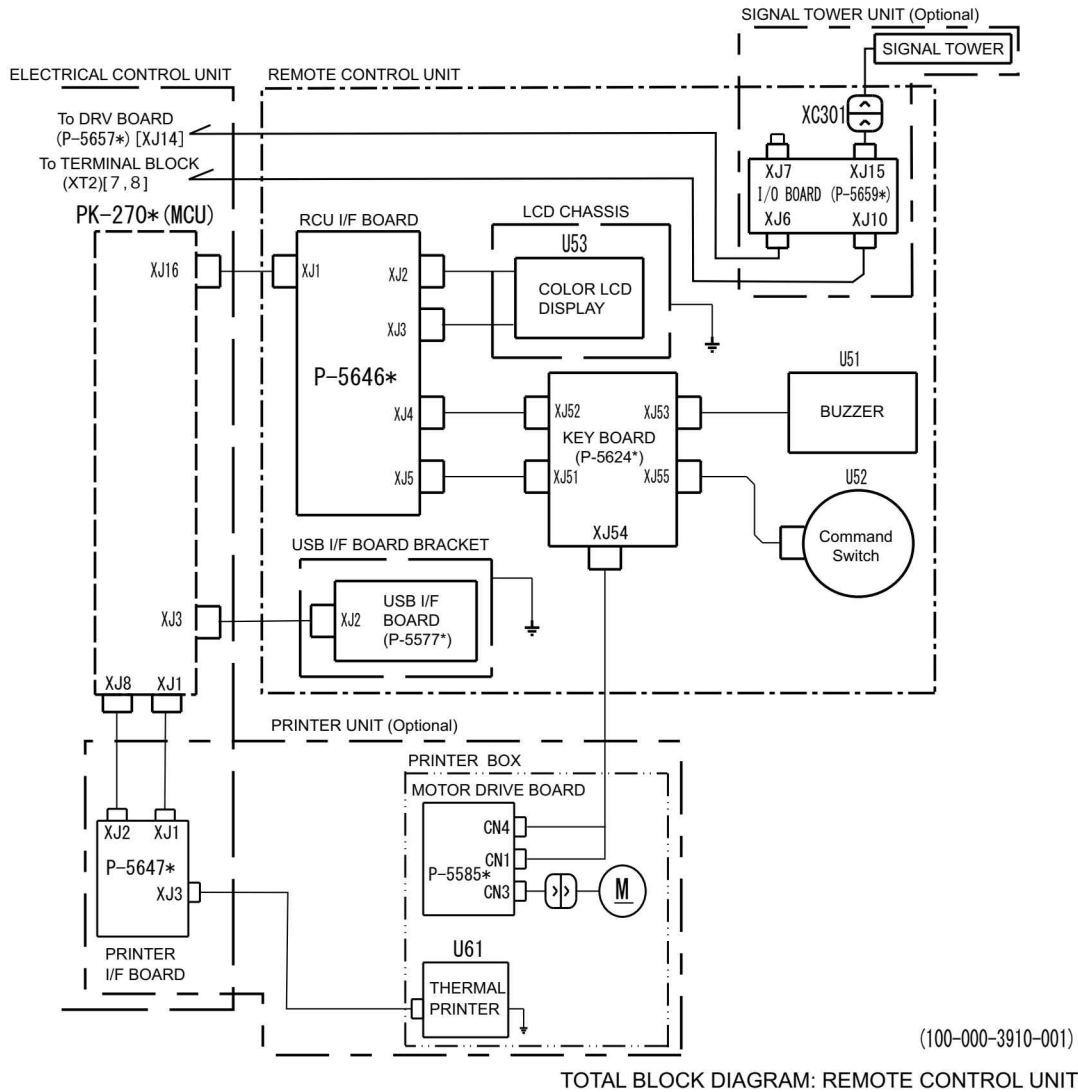
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4.2 Remote Control Unit

■ External View

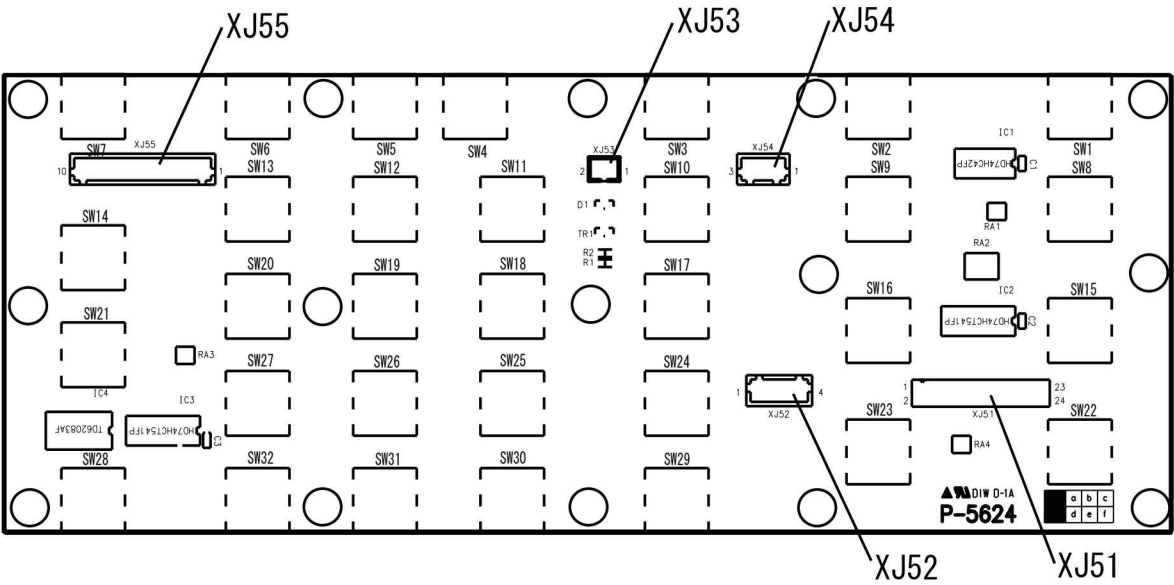


4.2.1 Block Diagram



4.2.2 Key Board (P-5624*)

■ External View



■ Functional Descriptions

- 1) Key input control
- 2) Signal output to RCU I/F board
- 3) Signal output to buzzer
- 4) Command dial input control
- 5) Printer paper wind-up motor control (optional)

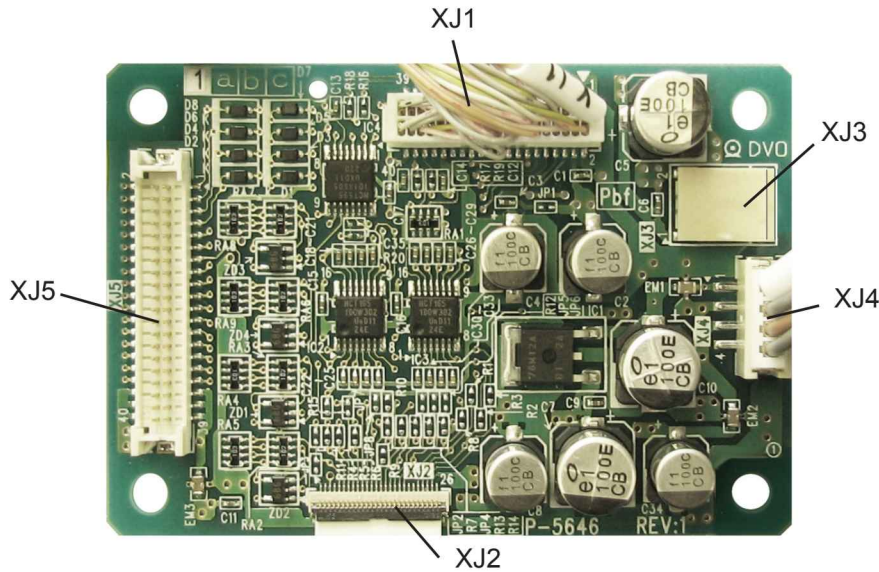
■ Connector usage classification

CONNECTOR No.		FUNCTION
XJ51	24P	Key input, LED, Buzzer ON signal
XJ52	4P	Power (12V, 5V)
XJ53	2P	Buzzer
XJ54	3P	To Winding Motor Drive Board (option)
XJ55	10P	Command dial

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4.2.3 RCU I/F board (P-5646*)

■ External View



■ Functional Descriptions

- 1) Display control of LCD unit
- 2) Communication to MCU board (PK-270*)
- 3) Power supply to backlight
- 4) Interface with a key board

■ Connector usage classification

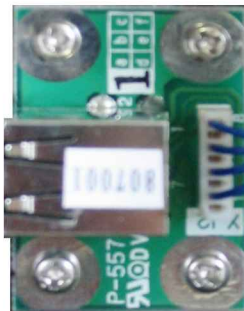
CONNECTOR No.		FUNCTION
XJ1	42P	Communication to MCU board (PK-270*)
XJ2	28P	LCD Control signal
XJ3	2P	LCD Backlight power supply
XJ4	4P	Power output to Key board (12 V, 5 V)
XJ5	40P	Key board interface

(2) XJ1: Communication signal to MCU board (PK-270*)

CONNECTOR No.	PIN No.	FUNCTION
X J 1	1	AVSS
	2	AVSS
	3	AVDD
	4	AVDD
	5	AVSS
	6	VR
	7	AVSS
	8	VG
	9	AVSS
	10	VB
	11	AVSS
	12	VSYN
	13	CKV
	14	AVSS
	15	CEV
	16	AVSS
	17	HSYN
	18	AVSS
	19	CPH
	20	CEH
	21	VCOM1
	22	VCOM2
	23	VGL
	24	VGH
	25	GND
	26	GND
	27	VLED
	28	GND2
	29	NC
	30	BUZZER
	31	VBR
	32	VCC
	33	VCC
	34	TXD
	35	RXD
	36	SO_DT
	37	SO_CK
	38	SI_CK
	39	SI_DT
	40	GND
	41	GND
	42	GND

4.2.4 USB Interface Board (P-5577*)

■ External View

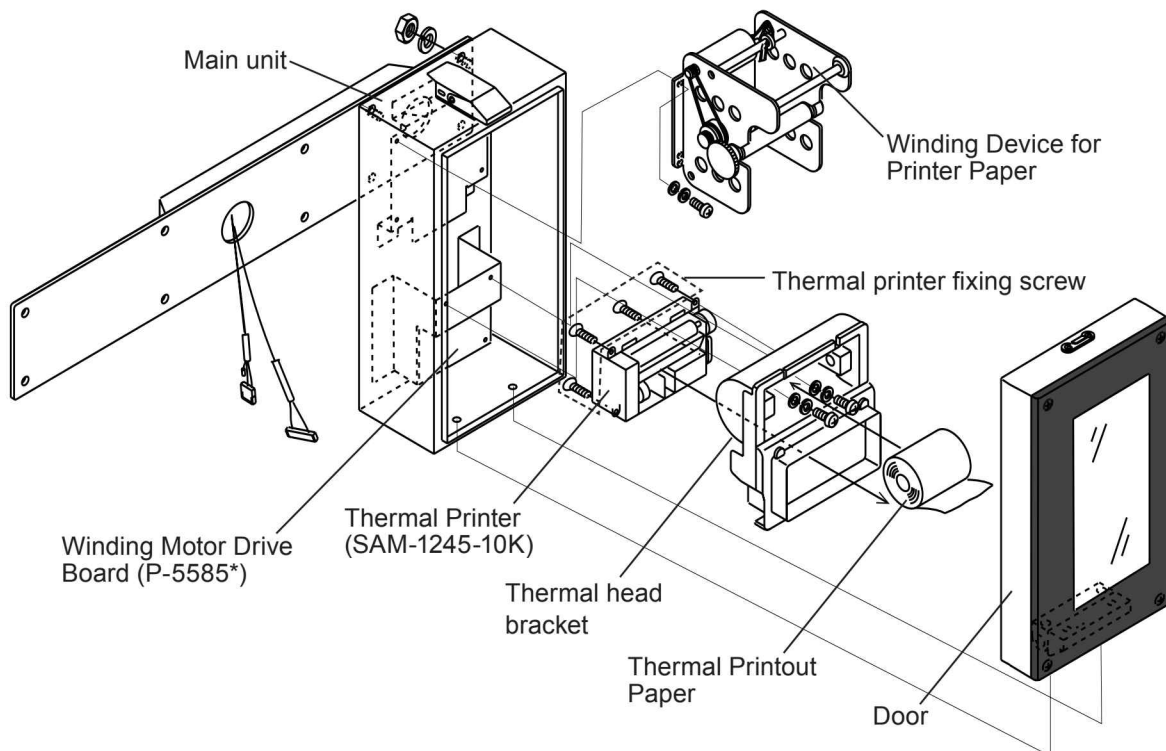


■ Board Function

- Connecting USB memory

4.3 Printer Unit (optional)

■ Appearance of Printer Unit

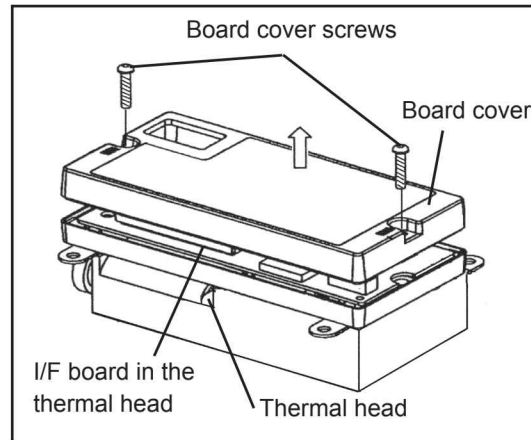


■ Unit Function

- (1) Printing statistical calculation results
- (2) Printing preset/set values

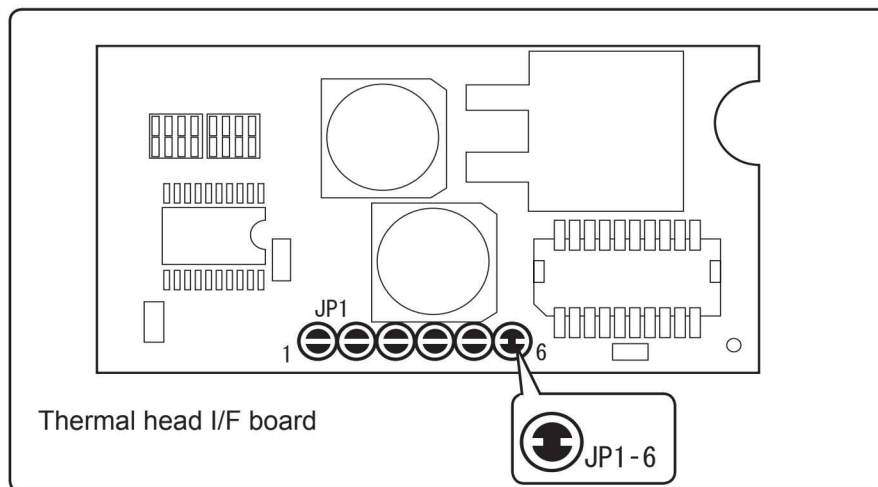
4.3.1 Thermal Printer (SAM-1245-10K)

■ Appearance of Thermal Printer



■ Replacement of Thermal Printer

1. Open the door of printer unit.
2. Remove the roll printing paper.
3. Remove two screws of the bracket (Plastic-made) fixing the thermal printer, and remove the bracket.
4. Turn the bracket upside down, and disconnect the connector connected to the thermal printer.
5. Disconnect three screws fixing the bracket to the thermal printer, and remove the printer.
6. Prepare a new thermal printer. Disconnect two mounting screws for board cover, and remove the board cover.
7. Perform the function selection setting of the thermal printer. Short-circuit the soldering jumper JP1-6 on the board shown below.
8. Install the thermal printer by the reverse procedure described from the step 1 to 7 above.



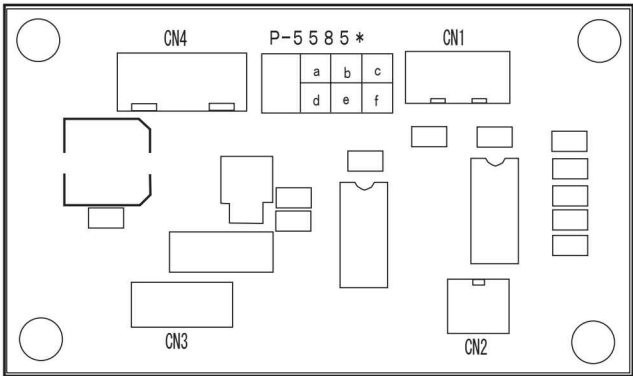
NOTE

In the new thermal printer, the soldering jumpers JP1-1 to JP1-6 are in the open state; be sure to perform the short-circuit between JP1-6. Otherwise, a correct operation cannot be carried out.

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4.3.1.1 Winding Motor Drive Board (P-5685*) (optional)

■ Board Location

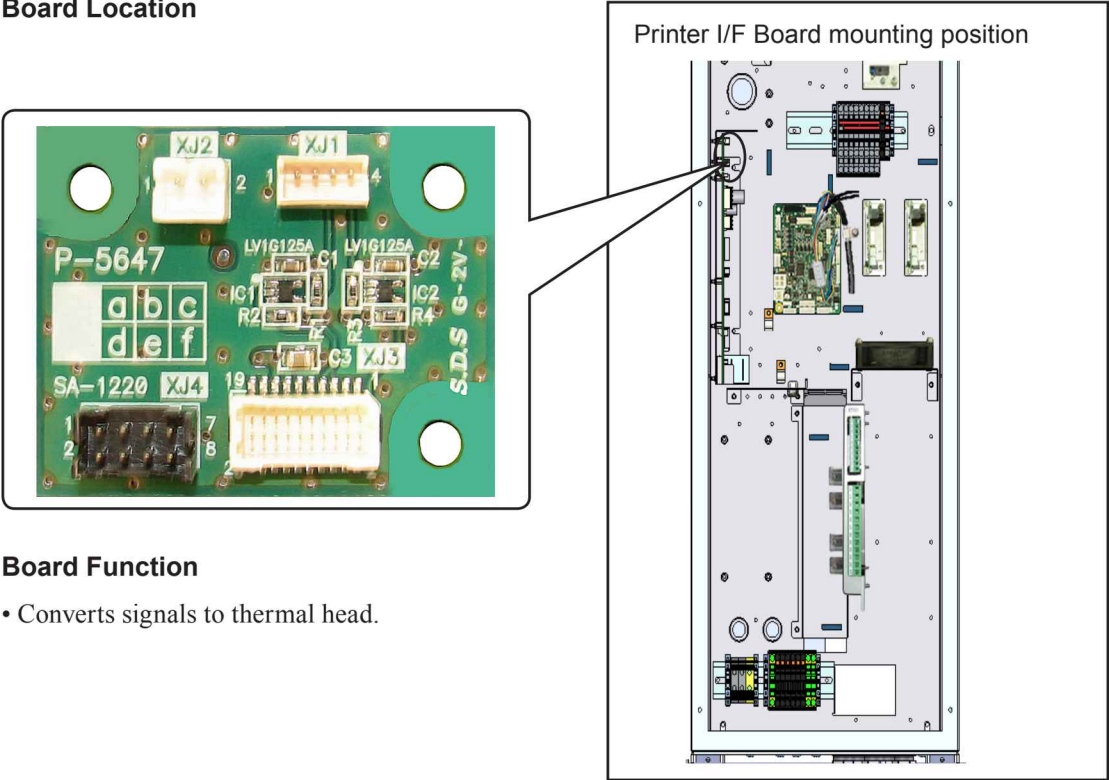


■ Board Function

- Control of Winding Motor

4.3.1.2 Printer I/F Board (P-5647*) (optional)

■ Board Location



■ Board Function

- Converts signals to thermal head.

■ Connector usage classification

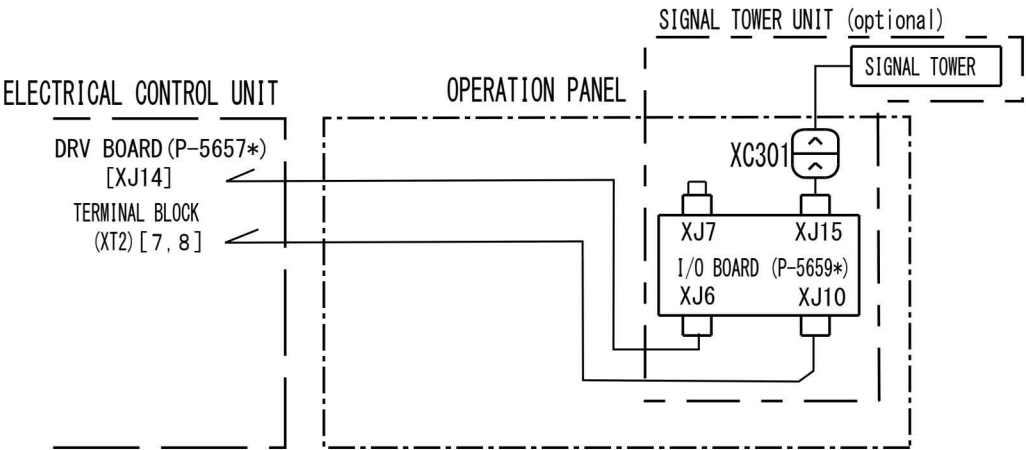
CONNECTOR No.		FUNCTION
XJ1	8P	Communication signal with DRV board (P-5657*)
XJ2	2P	Power supply input (+24 VDC)
XJ3	20P	Signal tower control signal

4.4 Signal tower unit (optional)

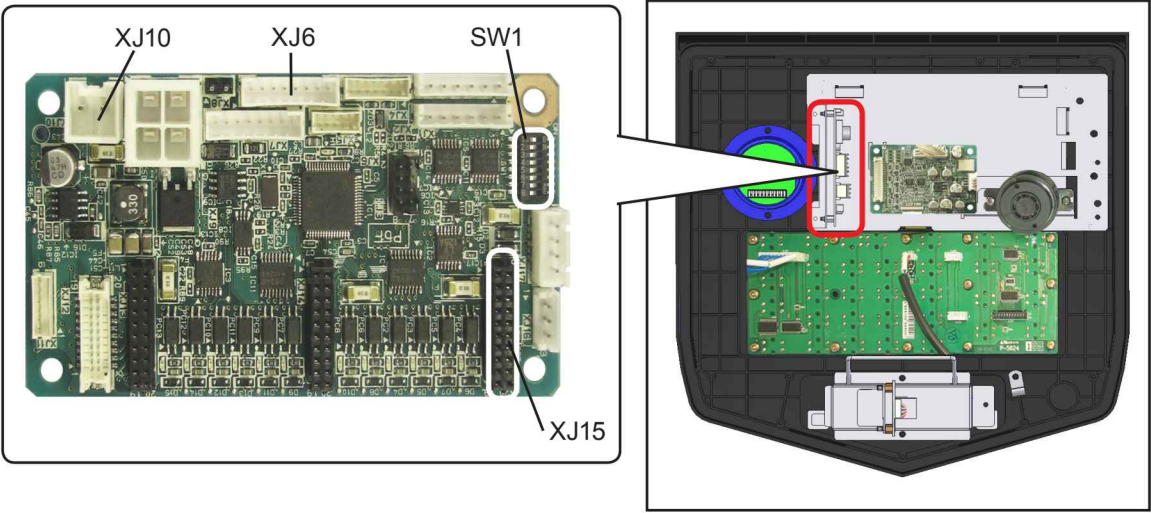
■ External View and location



■ Wiring Diagram



■ External View and location of I/O board (P-5659*)



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■ Board Function

- Output signal to signal tower
- Communication with DRV board (P-5657*)

■ DIP SW1 default setting

SW1-	FUNCTION		DEFAULT
1	Board ID setting 1	Always OFF at GN	OFF
2	Board ID setting 2	Always ON at GN	ON
3	Not assigned	Always OFF at GN	OFF
4	Not assigned	Always OFF at GN	OFF
5	Not assigned	Always OFF at GN	OFF
6	Not assigned	Always OFF at GN	OFF
7	Not assigned	Always OFF at GN	OFF
8	Boot Loader	OFF : APPLICATION START ON : BOOT LOARDEr START	OFF

NOTE

- If the power is shut down during software install, or failed software installation, set the No.8 of SW1 to ON then try installation software.
- After installation of software, do not fail to return the SW1-No.8 to OFF.

■ Connector usage classification

CONNECTOR No.		FUNCTION
XJ1	8P	Communication signal with DRV board (P-5657*)
XJ2	2P	Power supply input (+24 VDC)
XJ3	20P	Signal tower control signal

■ Program update after board replacement (installation) [Additional function]

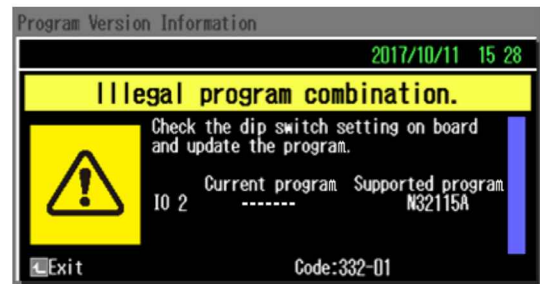
(Software version: Ver.3_5300J or later)

(MCU: N32110H, DRV:N3211E, ADC:32112, REJ:32114C, IO:32115A or later)

1. Set the switches DIP-SW1-1, and 3 to 7 on the board to replace to OFF, and set DIP-SW1-2, 8 to ON, and mount the board.

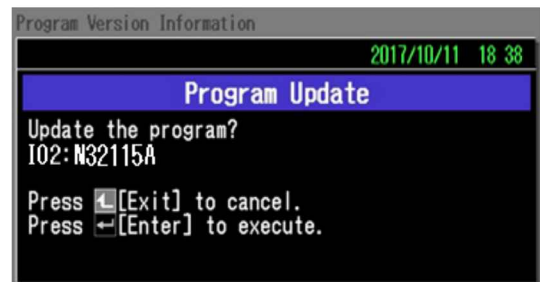
2. Turn ON the power switch.

- The Program Version Information screen for warning appears.



3. Press key.

- The Program Update screen to confirm the update (installation) appears.



4. Press key to perform the program update (installation).

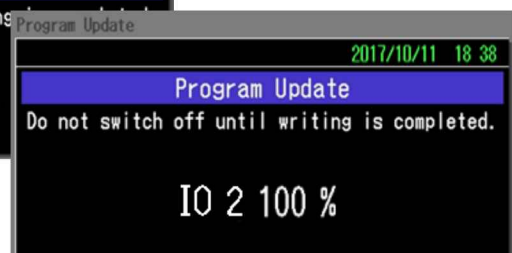
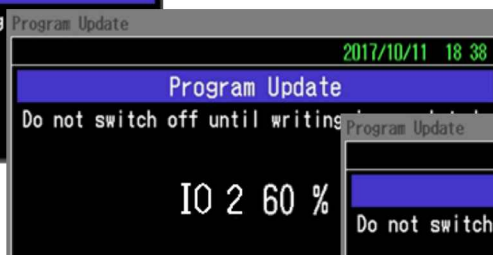
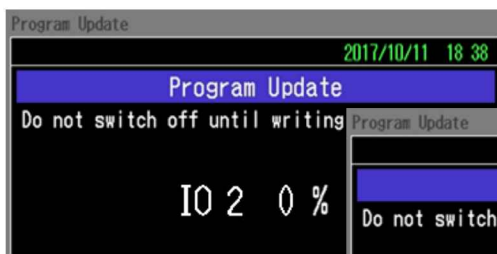
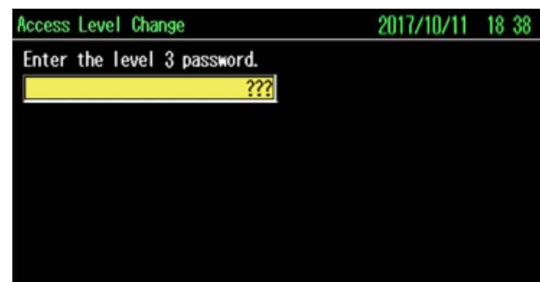
- The Access Level Change screen is displayed.
- Input the level 3 password.



Contact to ISHIDA agent for the password required in the level 3. "???" is displayed on the password displayed screen for security reasons.

5. Press key.

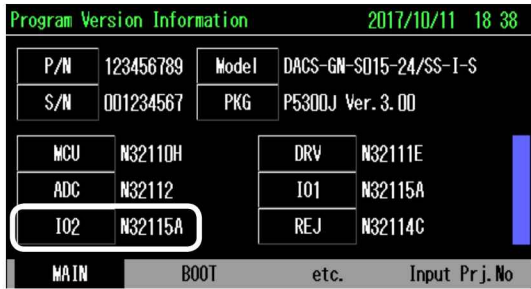
- The program update (installation) starts.



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4 Electrical Unit Configuration

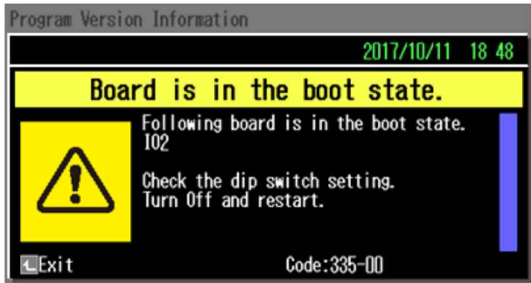
- 6. A message "Completed. Turn off the power and restart." appears. As indicated, turn off the power switch.
- 7. Return the switches DIP-SW1 to 8 on the board to the OFF state.
- 8. Turn ON the power switch again.
 - Point the cursor to [Setup Menu] - [Program Version Information], and press  key.
- 9. Check that the corresponding program No. is displayed on the Program Version Information screen.



■ Function to detect the boot state of the board program [additional function]

This function detects that the board program has started in the boot state (in which the program update is ready). In the boot state, normal operation is not available. There are two causes for the boot operation of the board program:

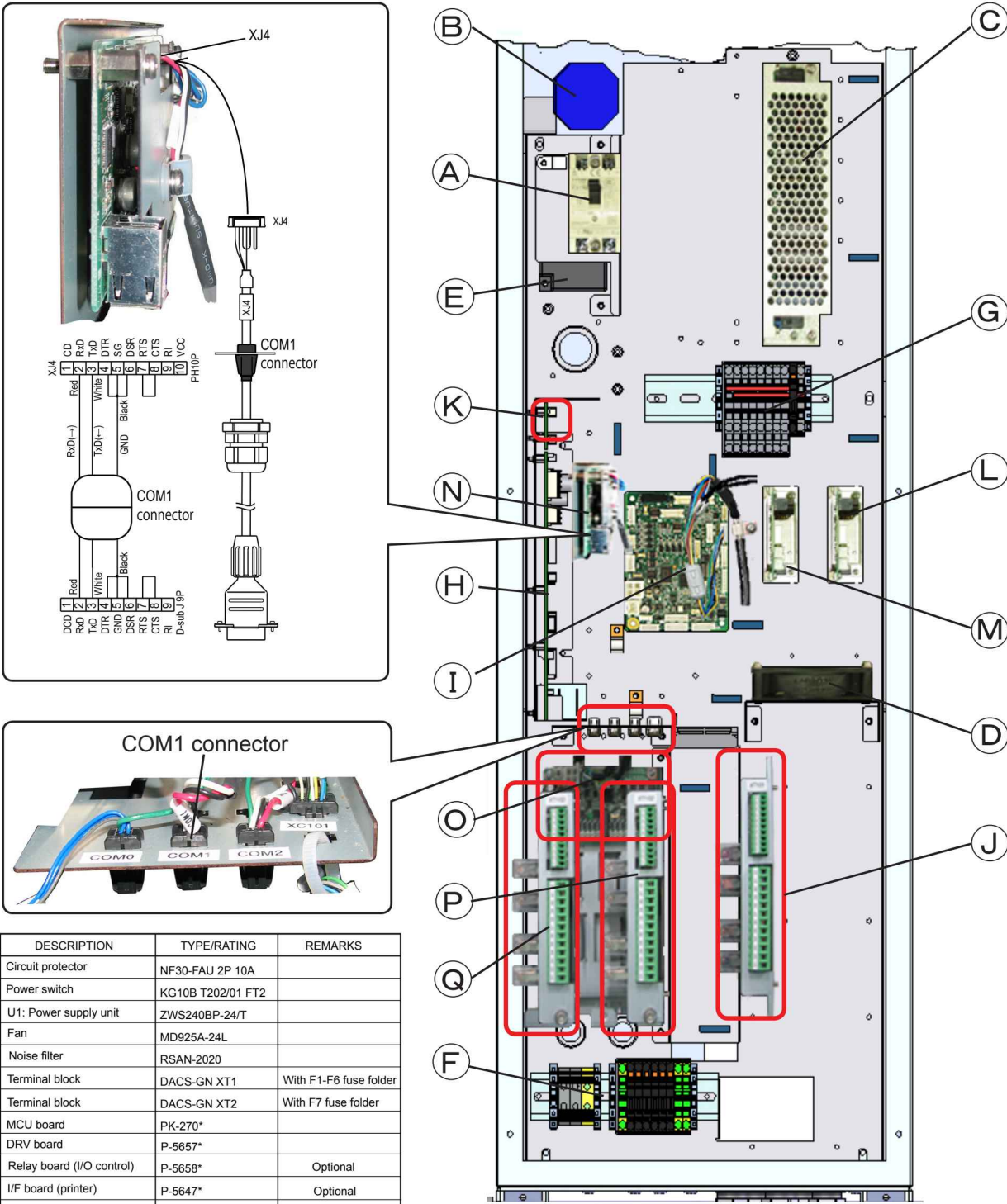
- 1) After downloading the program, power is turned ON soon after OFF.
(The residual voltage of the switching power supply disables the power OFF.)
- 2) The switches DIP-SW1 to 8 on the board are remained to be ON.



4.5 Electrical Unit

4.5.1 Parts Location of Electrical Unit

■ S015-XX/SS-I-S

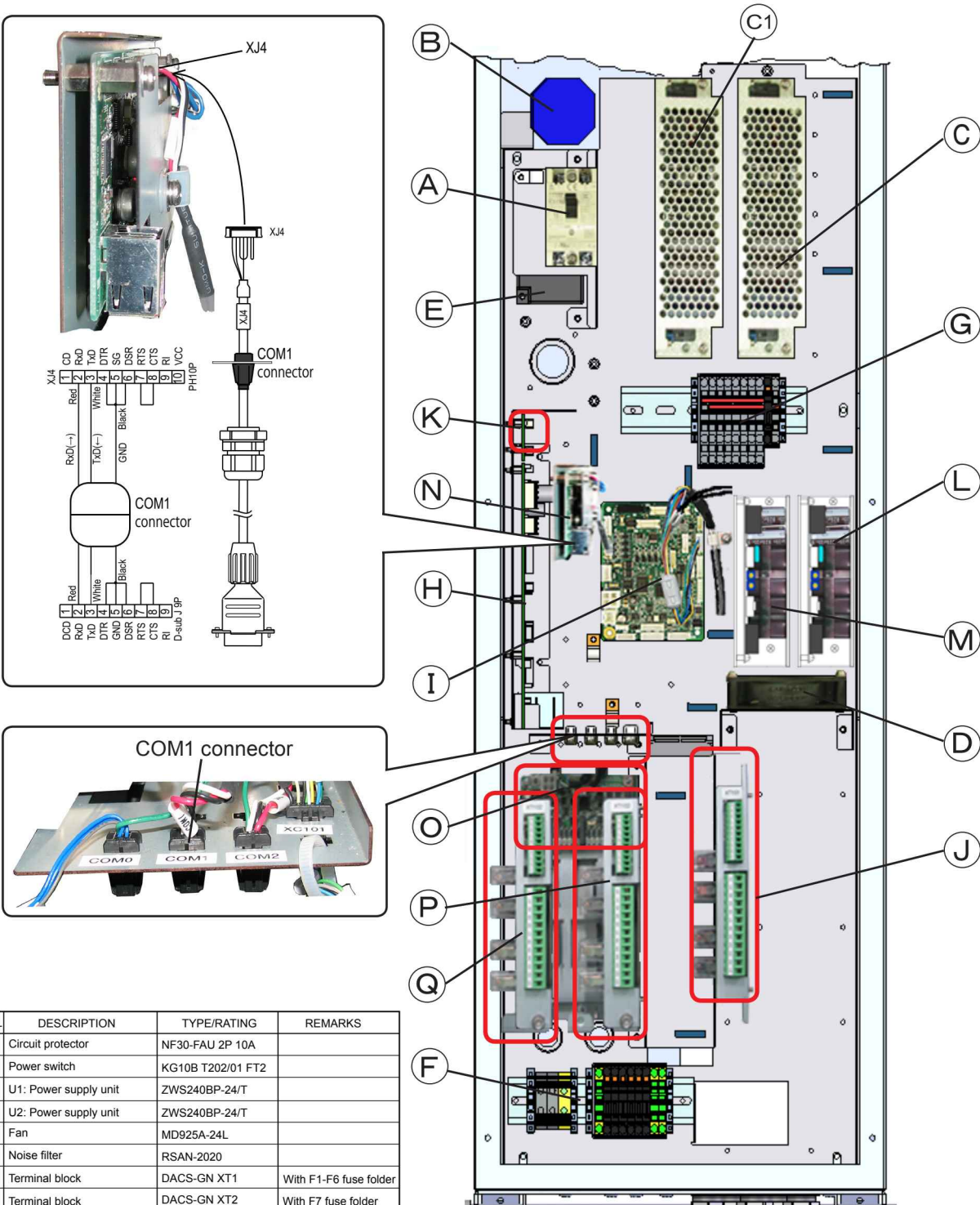


SYMBOL	DESCRIPTION	TYPE/RATING	REMARKS
A	Circuit protector	NF30-FAU 2P 10A	
B	Power switch	KG10B T202/01 FT2	
C	U1: Power supply unit	ZWS240BP-24/T	
D	Fan	MD925A-24L	
E	Noise filter	RSAN-2020	
F	Terminal block	DACS-GN XT1	With F1-F6 fuse folder
G	Terminal block	DACS-GN XT2	With F7 fuse folder
H	MCU board	PK-270*	
I	DRV board	P-5657*	
J	Relay board (I/O control)	P-5658*	Optional
K	I/F board (printer)	P-5647*	Optional
L	U3 (IFC) motor drive board	50W/100W	
M	U3 (WTC) motor drive board	30W/50W/100W	
N	USB serial board	P-1022*	Optional
O	I/O board	P-5659*	Optional
P	Relay board (Optional Input/Output signal control)	P-5658* (XT102)	Optional
Q	Relay board (Optional Input/Output signal control)	P-5658* (XT104)	Optional

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4 Electrical Unit Configuration

■ S060-XX/SS-I-S



SYMBOL	DESCRIPTION	TYPE/RATING	REMARKS
A	Circuit protector	NF30-FAU 2P 10A	
B	Power switch	KG10B T202/01 FT2	
C	U1: Power supply unit	ZWS240BP-24/T	
C1	U2: Power supply unit	ZWS240BP-24/T	
D	Fan	MD925A-24L	
E	Noise filter	RSAN-2020	
F	Terminal block	DACS-GN XT1	With F1-F6 fuse folder
G	Terminal block	DACS-GN XT2	With F7 fuse folder
H	MCU board	PK-270*	
I	DRV board	P-5657*	
J	Relay board (I/O control)	P-5658*	Optional
K	I/F board (printer)	P-5647*	Optional
L	U3 (IFC) motor drive board	50W/100W	
M	U3 (WTC) motor drive board	30W/50W/100W	
N	USB serial board	P-1022*	Optional
O	I/O board	P-5659*	Optional
P	Relay board (I/O control of optional)	P-5658* (XT102)	Optional
Q	Relay board (I/O control of optional)	P-5658* (XT104)	Optional

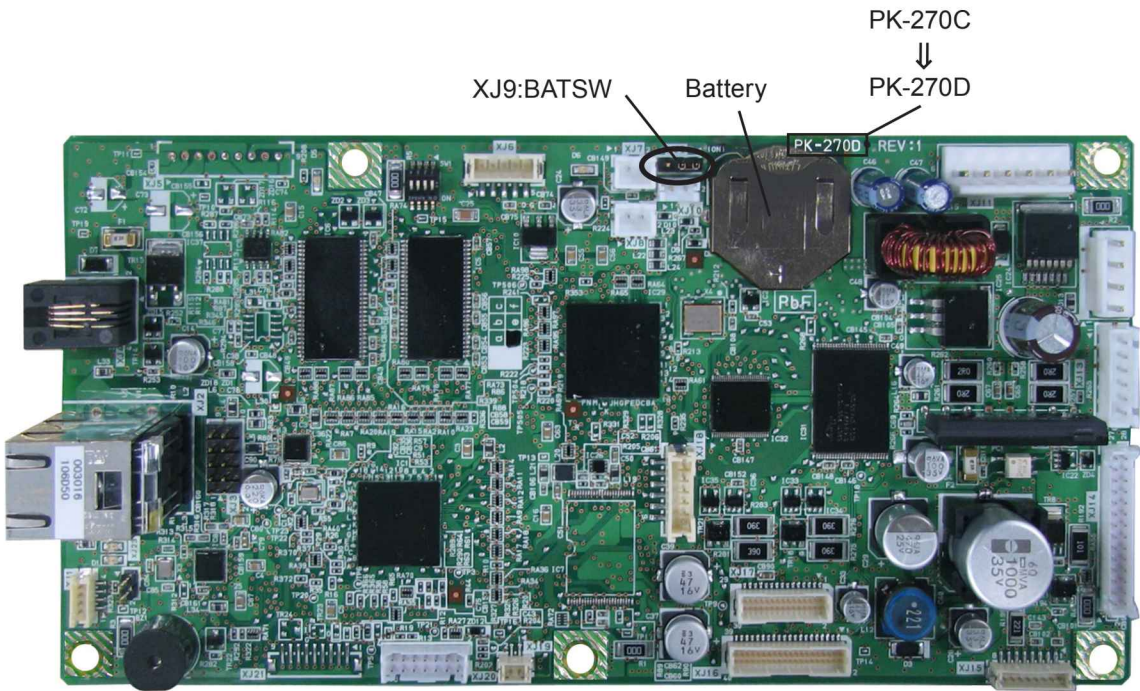
4.5.2 MCU Board (PK-270*→PK-270D)

■ Replacing MCU board

To the new MCU board (PK-270D), be sure to install the software for the new MCU board (P5510*).

CAUTION If the previous software (P5020*) is installed to the new MCU board (PK-270D) by mistake, the new MCU board (PK-270D) may be damaged. Be sure to install the correct software.

■ External View



■ Board Function

- 1) Communication control with USB interface board (P-5577*)
- 2) Communication control with DRV board (P-5657*)(RS485)
- 3) Communication control with RCU I/F board (P-5646*)
- 4) Memory of the statistical processing, reservation contents and common setting
- 5) Printer control (optional)

■ XJ9(BATSW) : Switch battery ON/OFF

BATSW			ON
	OFF	ON	Default

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CAUTION BATSW after changing MCU board: Never fail to set the battery ON/OFF switch pin “ON” when used.

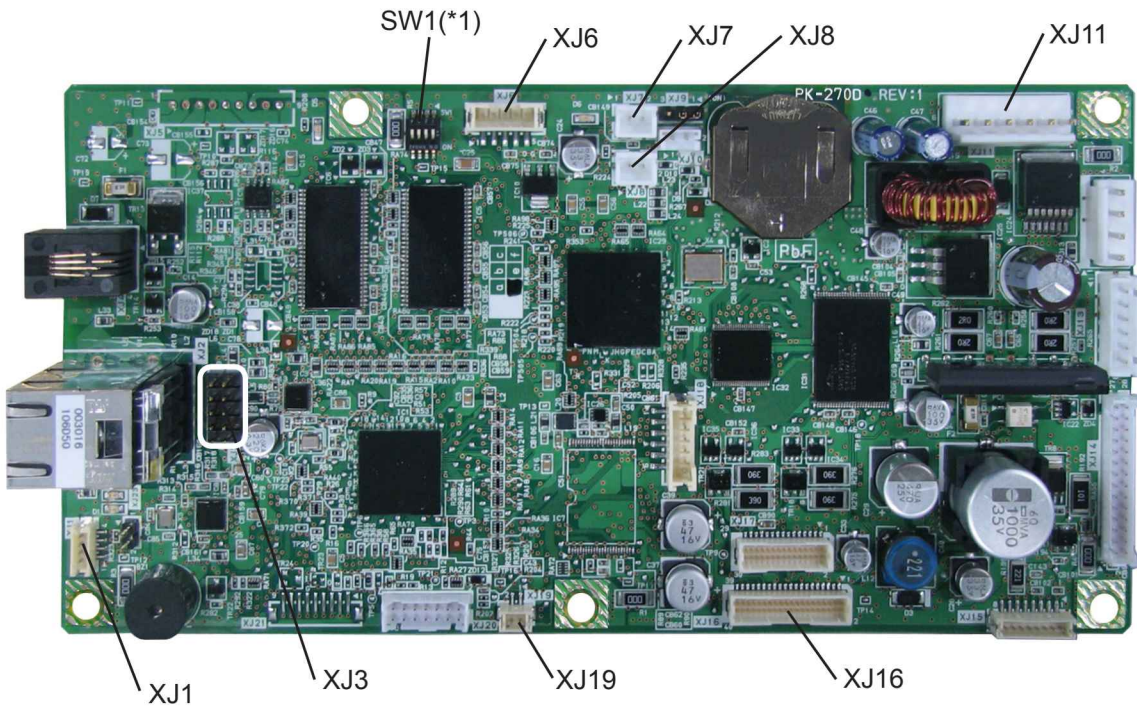
■ Battery

- Battery in use: Lithium battery (Storage battery) CR2450B SONY

CAUTION

- Risk of Explosion if replaced by an incorrective.
- If the battery is replaced with an incorrect one, it may result in a board malfunction.
- When replacing a battery, make sure that it is the same model or an equivalent.
- Dispose of used batteries after replacement.

■ Connector locations



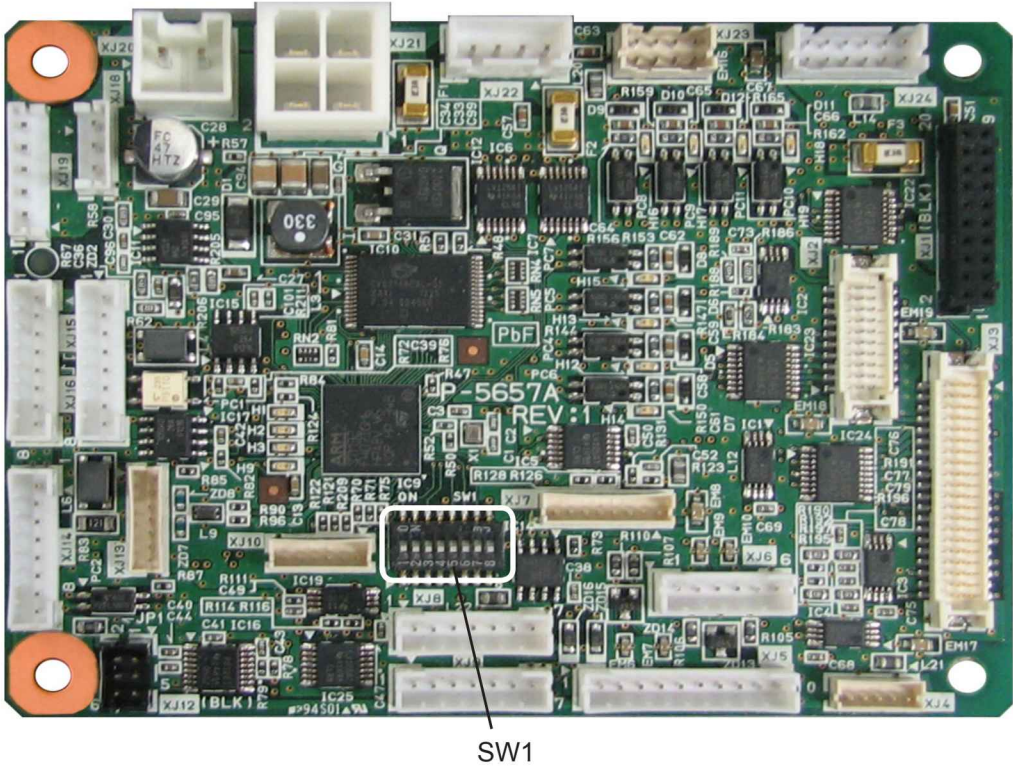
(*1) : Always set SW1 to OFF.

■ Connector usage classification

CONNECTOR No.		FUNCTION
XJ1	4P	Communication with Printer I/F board (P-5647*)(optional)
XJ3	10P	Communication with USB connector board (P-5577*)
XJ6	5P	Communication with DRV board (P-5657*) [RS485]
XJ7	2P	Power detection of DRV board (P-5657*)(+5 V)
XJ8	2P	Power supply to Printer I/F board (P-5647*)(+5 V)(optional)
XJ11	6P	Power input (+24 V)
XJ16	40P	Communication with RCU I/F board (P-5646*)
XJ19	2P	Communication ready signal with DRV board (P-5657*)

4.5.3 DRV Board (P-5657*)

■ External View



■ Board Function

- 1) Control of infeed/weigh/reject conveyor motors
- 2) Controls of contact point signals
- 3) Control of fan motor
- 4) Controls of weigh signal

■ Function of DIP-SW1

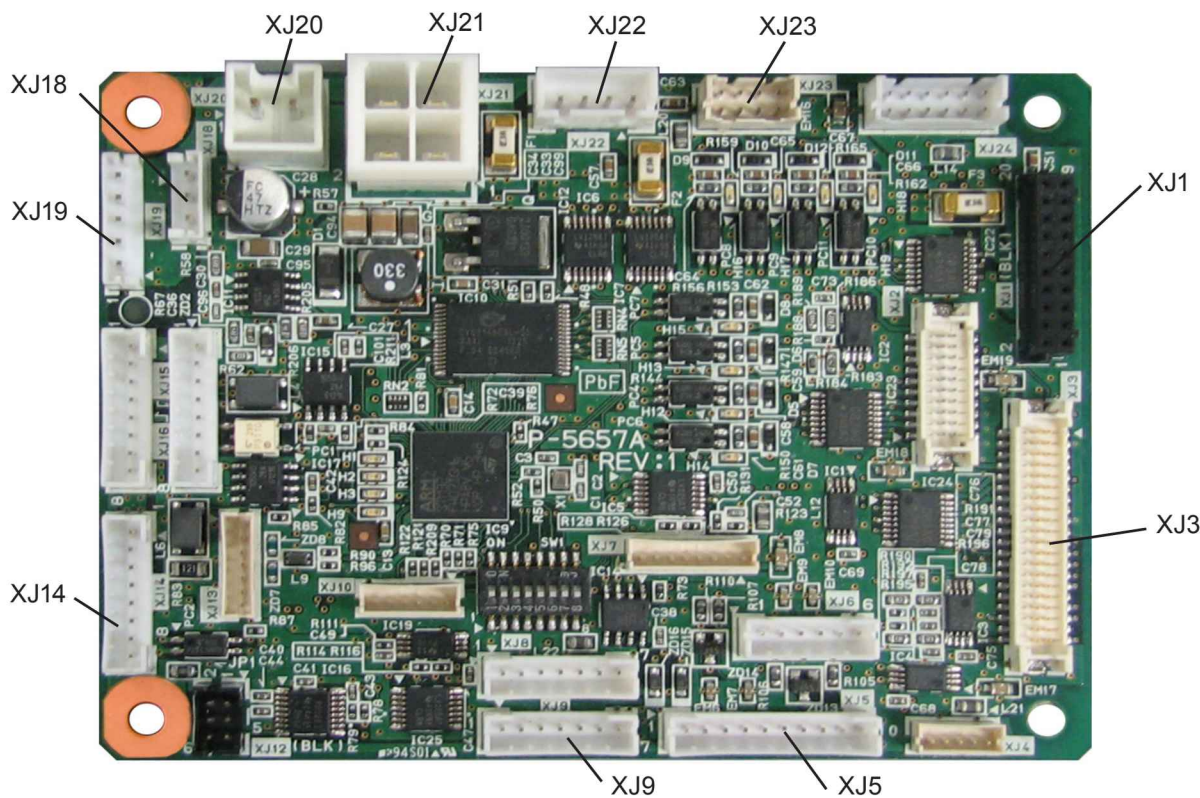
SW1-	FUNCTION		DEFAULT
1	Board ID setting 1	Always OFF at GN	OFF
2	Board ID setting 2	Always OFF at GN	OFF
3	Board ID setting 3	Always OFF at GN	OFF
4	Board ID setting 4	Always OFF at GN	OFF
5	Board ID setting 5	Always OFF at GN	OFF
6	Not assigned	Always OFF at GN	OFF
7	Not assigned	Always OFF at GN	OFF
8	Boot Loader	OFF : APPLICATION START ON : BOOT LOADER START	OFF

NOTE

If the power is shut down during software install, or failed software installation, set the No.8 of SW1 to ON then try installation software.
After installation of software, do not fail to return the SW1-No.8 to OFF.

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■ Connector Locations



■ Connector usage classification

CONNECTOR NO.		FUNCTION
XJ1	20P	Communication with Relay board (P-5658*)(optional)
XJ3	40P	U3/U4 motor drive board control
XJ5	10P	Weigh data from ADC board (P-5660*)
XJ9	5P	Communication with MCU board (PK-270*)(RS485)
XJ14	8P	Communication with I/O board (P-5659*)(optional)
XJ18	3P	Cooling fan (M2)
XJ20	2P	Power input (+24 V)
XJ21	4P	Power Source of integrated pneumatic rejector (+24V) (Short-circuit specification)
XJ22	4P	Rejector solenoid valve (L/R) output (+24 V) [For integrated pneumatic rejection]
XJ23	8P	Air pressure sensor, passed product sensor (optional)

■ Program update after board replacement (installation) [Additional function]

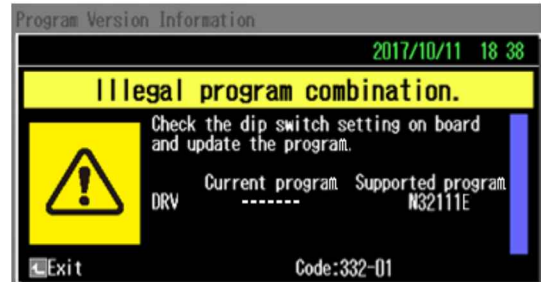
(Software version: Ver.3_5300J or later)

(MCU: N32110H, DRV:N3211E, ADC:32112, REJ:32114C, IO:32115A or later)

1. Set the switches DIP-SW1-1, and 1 to 7 on the board to replace to OFF, and set DIP-SW1- 8 to ON, and mount the board.

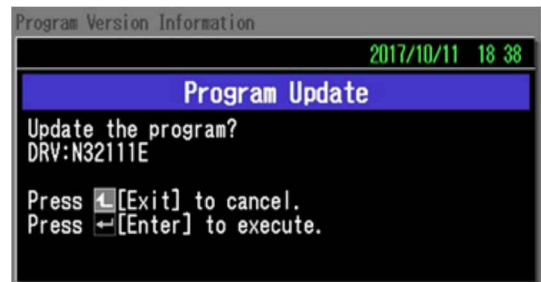
2. Turn ON the power switch.

- The Program Version Information screen for warning appears.



3. Press  key.

- The Program Update screen to confirm the update (installation) appears.



4. Press  key to perform the program update (installation).

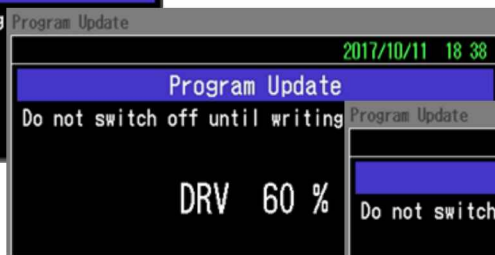
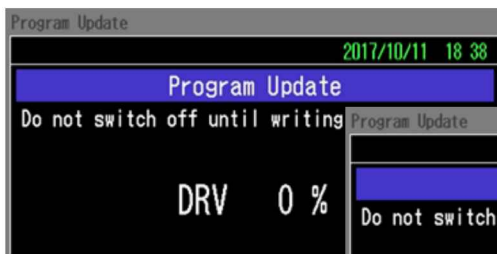
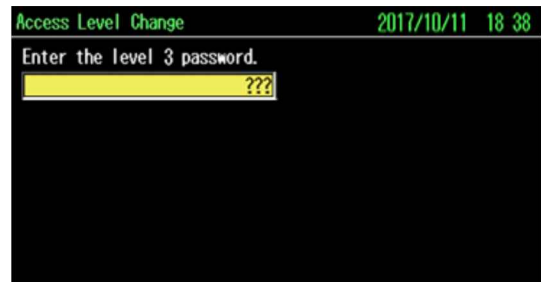
- The Access Level Change screen is displayed.
- Input the level 3 password.



Contact to ISHIDA agent for the password required in the level 3. "???" is displayed on the password displayed screen for security reasons.

5. Press  key.

- The program update (installation) starts.



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4 Electrical Unit Configuration

- 6. A message "Completed. Turn off the power and restart." appears. As indicated, turn off the power switch.
- 7. Return the switches DIP-SW1 to 8 on the board to the OFF state.



- 8. Turn ON the power switch again.
 - Point the cursor to [Setup Menu] - [Program Version Information], and press  key.



- 9. Check that the corresponding program No. is displayed on the Program Version Information screen.



■ Function to detect the boot state of the board program [additional function]

This function detects that the board program has started in the boot state (in which the program update is ready). In the boot state, normal operation is not available. There are two causes for the boot operation of the board program:

- 1) After downloading the program, power is turned ON soon after OFF.
(The residual voltage of the switching power supply disables the power OFF.)
- 2) The switches DIP-SW1 to 8 on the board are remained to be ON.



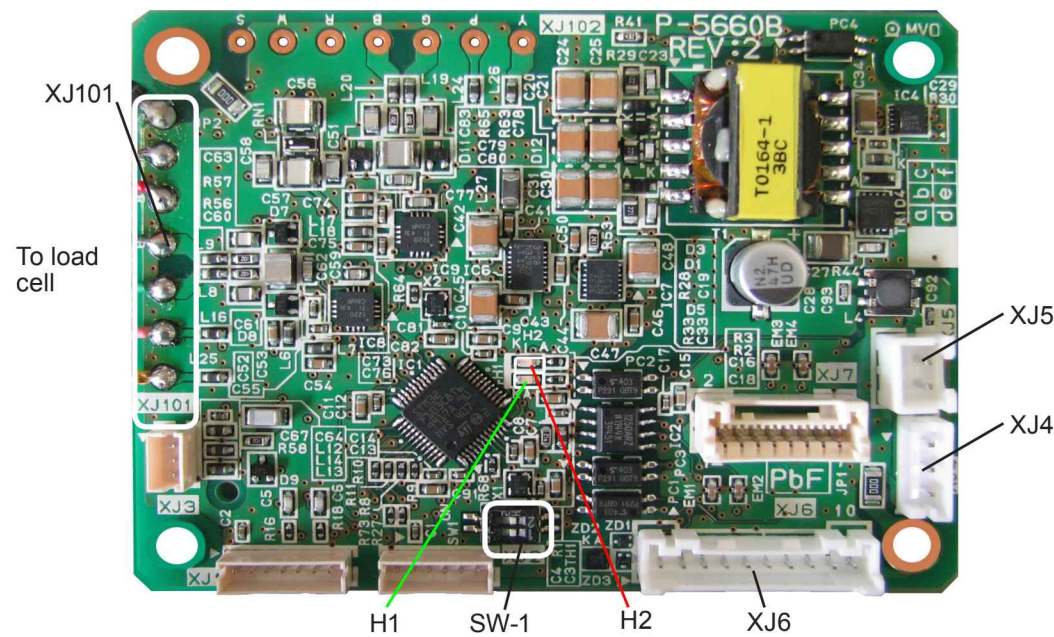
*When the DRV board boots up, the screen turns to "Communication Error (DRV)".

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4.5.4 ADC Board (P-5660*)

NOTE Replace the ADC board (P-5660*) along with the load cell. The load cell adjustments such as temperature compensation is performed by the ADC board (P5660*).

■ External View and Location



■ Board Function

- 1) Digitally converts the load cell analog signal
- 2) Controls the temperature compensation of the load cell
- 3) Sends the weight data signal to the DRV board through the digital filter
- 4) Relays photo sensor signal

■ Details of connector function

SW1-	FUNCTION		DEFAULT
1	ON: infeed side.	OFF: Weigh side	OFF
2	ON: Boot loader start	OFF: Application start	OFF

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■ Program update after board replacement (installation) [Additional function]

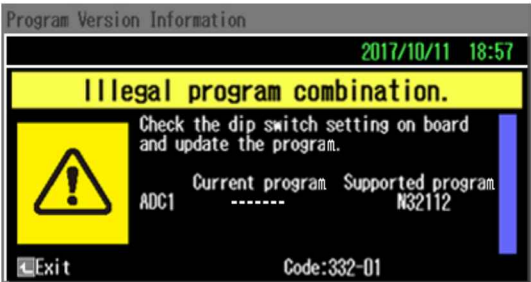
(Software version: Ver.3_5300J or later)

(MCU: N32110H, DRV:N3211E, ADC:32112, REJ:32114C, IO:32115A or later)

1. Set the switches DIP-SW1-1 on the board to replace to OFF, and set DIP-SW1-2 to ON, and mount the board.

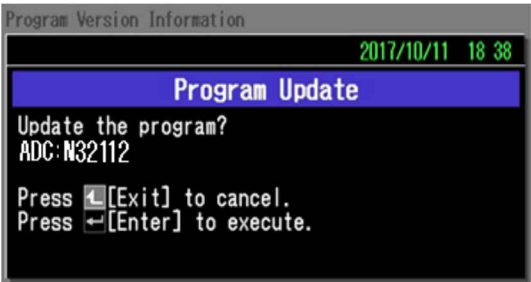
2. Turn ON the power switch.

- The Program Version Information screen for warning appears.



3. Press  key.

- The Program Update screen to confirm the update (installation) appears.



4. Press  key to perform the program update (installation).

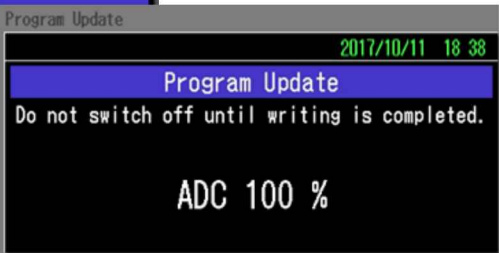
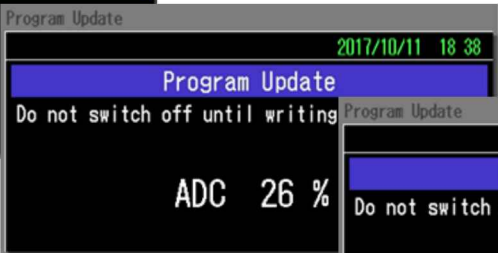
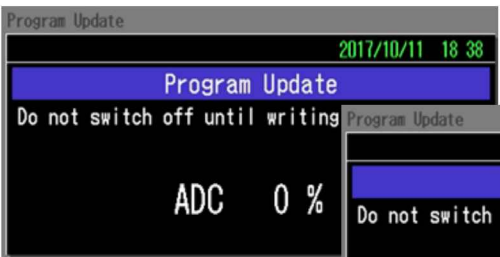
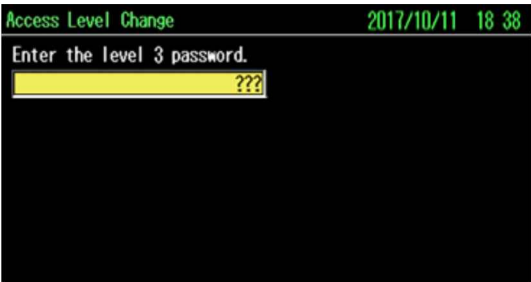
- The Access Level Change screen is displayed.
- Input the level 3 password.



Contact to ISHIDA agent for the password required in the level 3. "???" is displayed on the password displayed screen for security reasons.

5. Press  key.

- The program update (installation) starts.



6. A message "Completed. Turn off the power and restart." appears. As indicated, turn off the power switch.



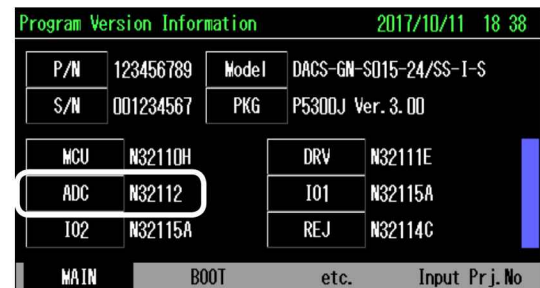
7. Return the switches DIP-SW1-2 on the board to the OFF state.

8. Turn ON the power switch again.

- Point the cursor to [Setup Menu] - [Program Version Information], and press key.



9. Check that the corresponding program No. is displayed on the Program Version Information screen.



■ Function to detect the boot state of the board program [additional function]

This function detects that the board program has started in the boot state (in which the program update is ready). In the boot state, normal operation is not available. There are two causes for the boot operation of the board program:

- After downloading the program, power is turned ON soon after OFF.
(The residual voltage of the switching power supply disables the power OFF.)
- The switches DIP-SW1 to 8 on the board are remained to be ON.



*When the ADC board boots up, the screen turns to "Communication Error (ADC)".

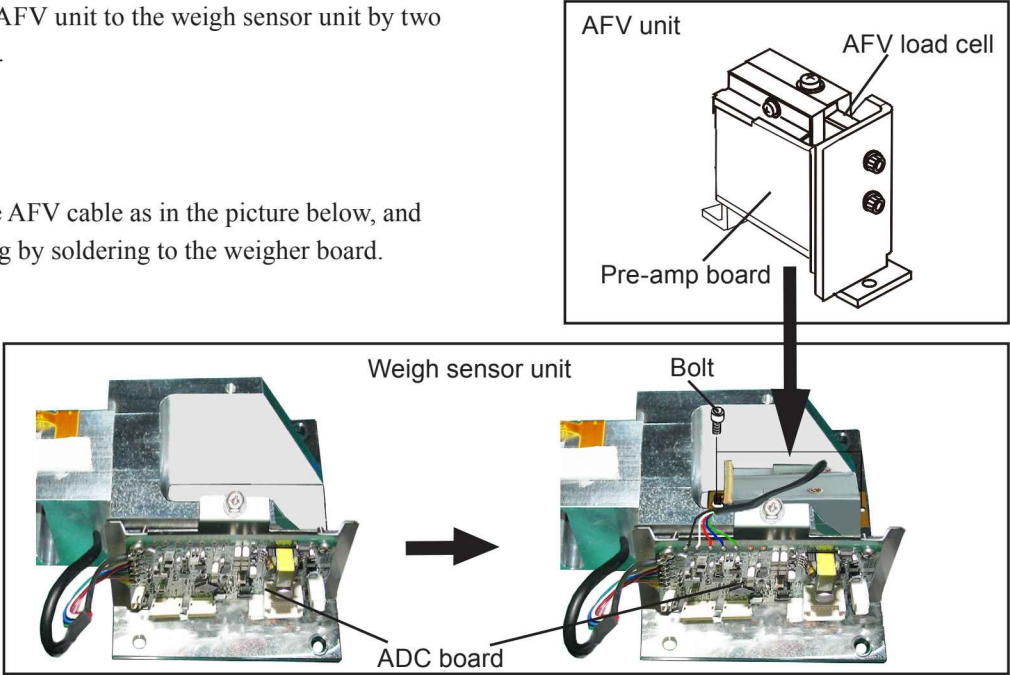
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4 Electrical Unit Configuration

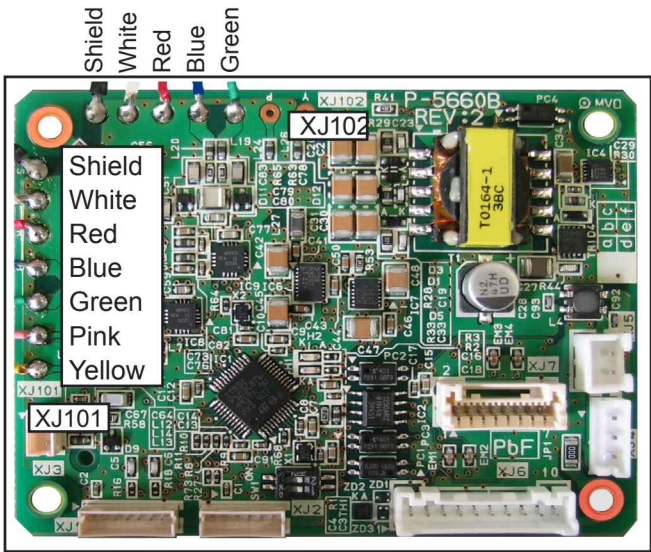
4.5.4.1 Mounting AFV unit (optional)

In mounting the AFV unit additionally, follow the step below.
Mount the AFV unit to the weigh sensor unit by two fixing bolts.

- 1. Arrange the AFV cable as in the picture below, and allow wiring by soldering to the weigher board.



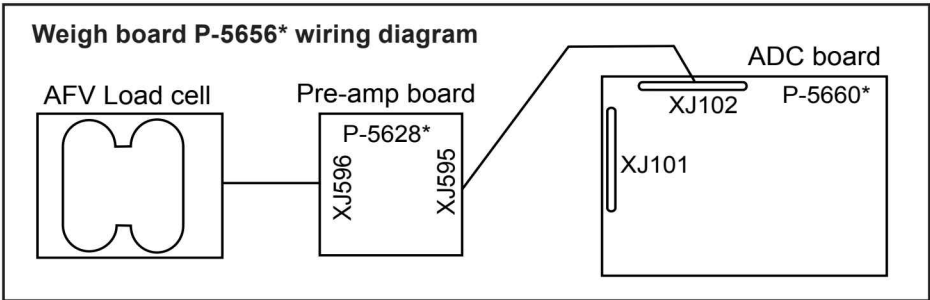
• Wiring arrangement of weigher board (P-5660*)



- 2. Connect lead wires of four-core shield cable coming from the AFV unit as above.



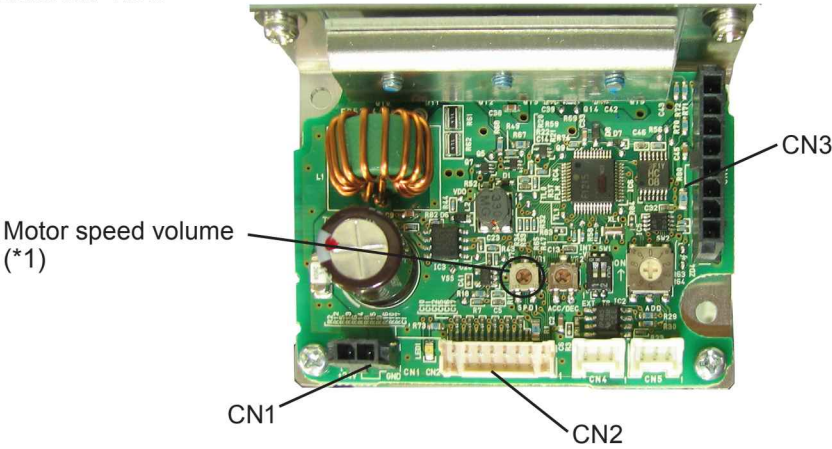
ADC board P-5656* wiring diagram



4.5.5 Motor Drive Board (U3, U4)

4.5.5.1 Motor Drive Board (30W/50W)

■ External View



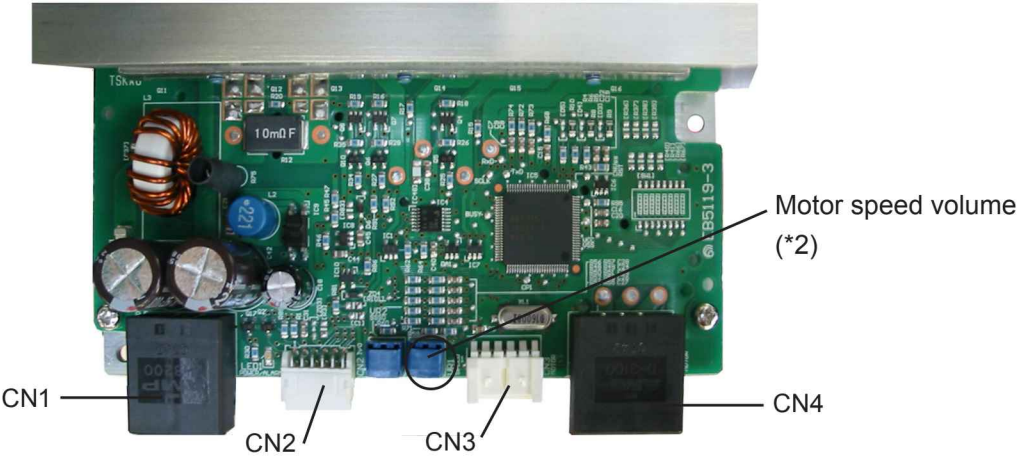
(*1): Set the volume at the new board to the same value to the previous board.

■ Connector usage classification

CONNECTOR No.		FUNCTION
CN1	2P	Power input (+24 V DC)
CN2	16P	Communication with DRV board (P-5657*)[XJ3]
CN3	8P	Motor control output (M2,M3)

4.5.5.2 Motor Drive Board (100W)

■ External View



(*2): Set the volume at the new board to the same value to the previous board.

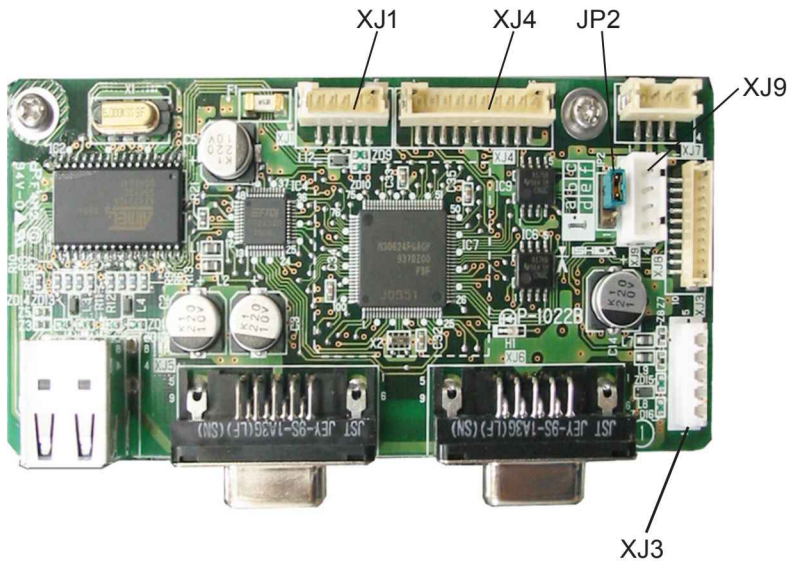
■ Connector usage classification

CONNECTOR NO.		FUNCTION
CN1	2P	Power input (+24 V DC)
CN2	12P	Communication with DRV board (P-5657*)[XJ3]
CN3	6P	Motor control output (M2, M3)
CN4	3P	Motor power supply (M2, M3)

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4.6 USB Serial Board (P-1022*) (Optional)

■ External View of USB Board (P-1022*)

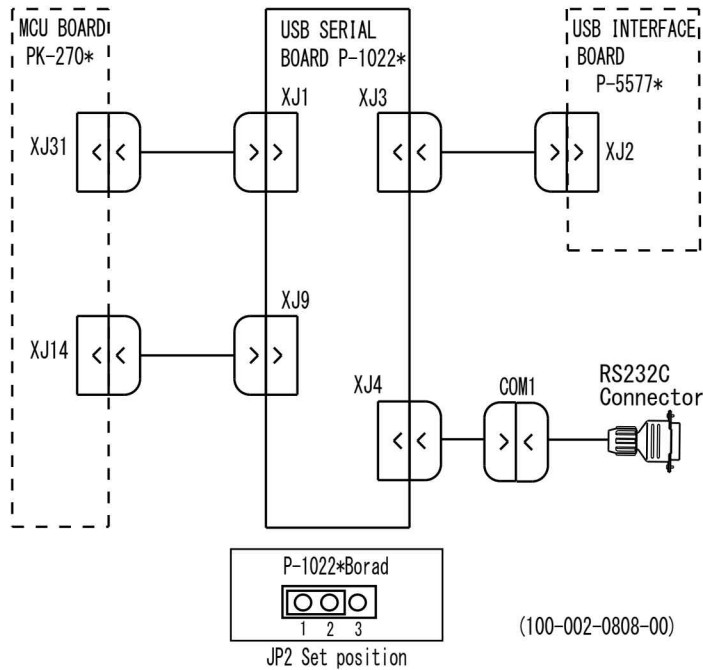


■ Board Functions

Outputs serial signal of preset, weigh data from RS232C

■ Block Diagram

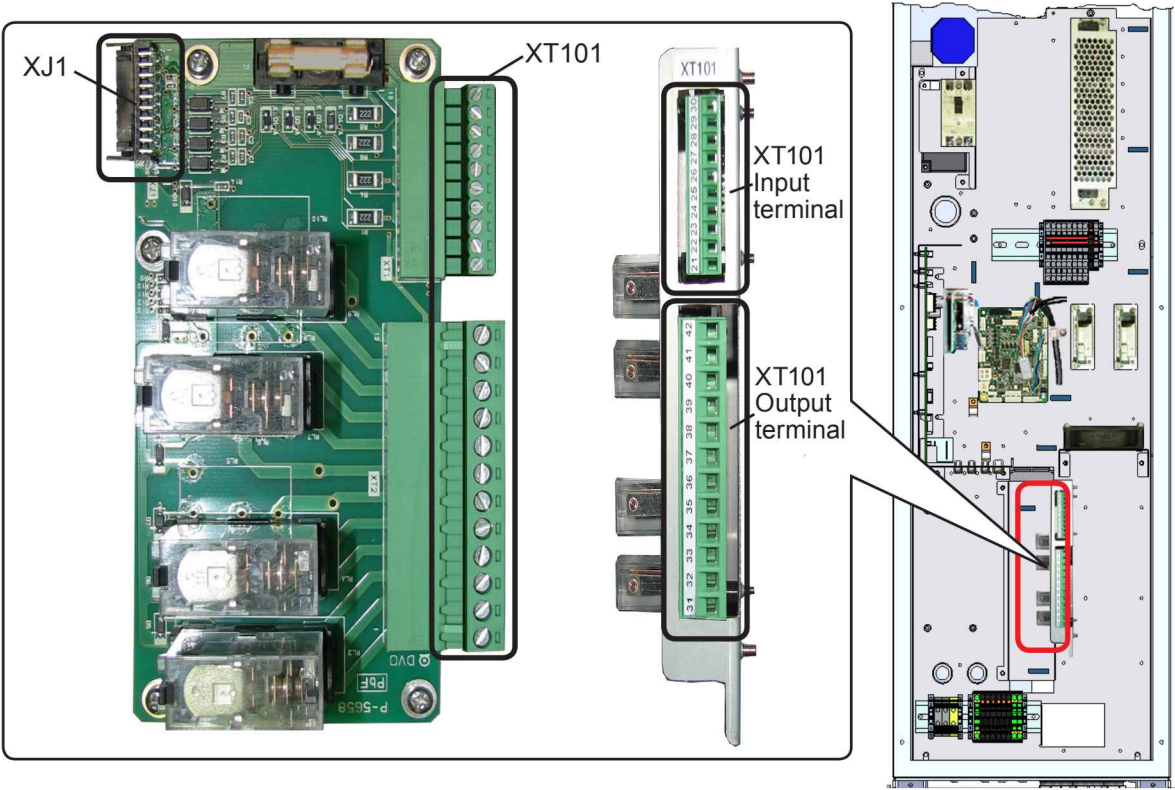
A [1 port]



4.7 Relay Terminal Unit (optional)

Input/output signals are optional. Adding a relay terminal unit can add signals of 4 ports for input and 4 ports for output, and to each of port the type of input/output signal can be selectively set via the remote control. The integrated metal detector (cosmo-one) integrated specification enables communication of input/output signals via the relay board.

■ Relay board (P-5658*) Location



■ Board Function

- Transmits external output signal
- Receives external input signal
- Communication with DRV board (P-5657*)

■ Connector usage classification

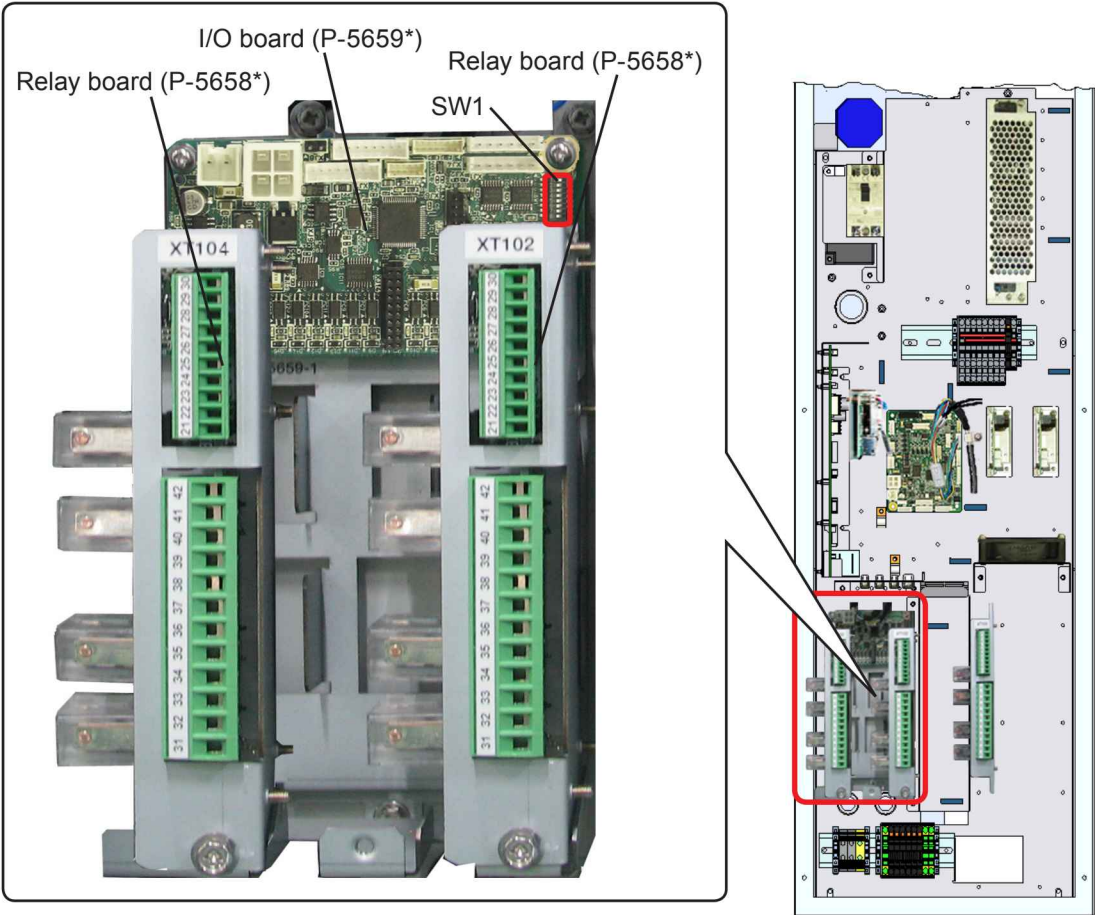
CONNECTOR		FUNCTION
XJ1	20P	Communication with DRV board (P-5657*)
XT101	10P	Input terminal
XT101	12P	Output terminal

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4.8 I/O unit (optional)

Adding the I/O board (P-5659*) can add two relay boards (P-5658*), which means up to eight ports of signals for input, eight ports of signals for output can be added. Select suitable Input/output signals for each port by selecting them from the remote control. The input/output signals for the other interlock devices are communicated via the relay board (P-5658*).

■ Parts Location of I/O Unit



■ Board Function

- Transmits external output signal
- Receives external input signal
- Communication with DRV board (P-5657*)

■ DIP SW1 default setting

SW1-	FUNCTION		DEFAULT
1	Board ID setting 1	Always ON at GN	ON
2	Board ID setting 2	Always OFF at GN	OFF
3	Not assigned	Always OFF at GN	OFF
4	Not assigned	Always OFF at GN	OFF
5	Not assigned	Always OFF at GN	OFF
6	Not assigned	Always OFF at GN	OFF
7	Not assigned	Always OFF at GN	OFF
8	Boot Loader	OFF : APPLICATION START ON : BOOT LOADER START	OFF

NOTE

If the power is shut down during software install, or failed software installation, set the No.8 of SW1 to ON then try installation software.

After installation of software, do not fail to return the SW1-No.8 to OFF.

■ Connector usage classification

CONNECTOR No.		FUNCTION
XJ6	3P	Communication with DRV board (P-5657*)
XJ10	2P	Power supply (+24V DC)
XJ13	20P	Communication with Relay board (P-5658*) [Adding of input/output signals]
XJ15	20P	Communication with Relay board (P-5658*) [Adding of input/output signals]

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■ Program update after board replacement (installation) [Additional function]

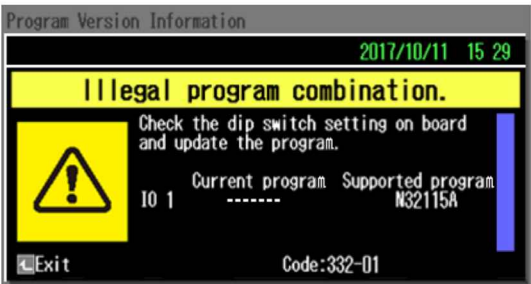
(Software version: Ver.3_5300J or later)

(MCU: N32110H, DRV:N3211E, ADC:32112, REJ:32114C, IO:32115A or later)

1. Set the switches DIP-SW1-2, and 2 to 7 on the board to replace to OFF, and set DIP-SW1-1, 8 to ON, and mount the board.

2. Turn ON the power switch.

- The Program Version Information screen for warning appears.



3. Press  key.

- The Program Update screen to confirm the update (installation) appears.

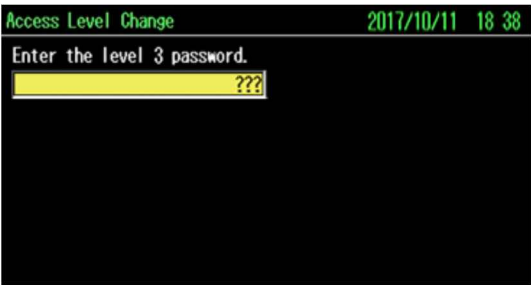


4. Press  key to perform the program update (installation).

- The Access Level Change screen is displayed.
- Input the level 3 password.

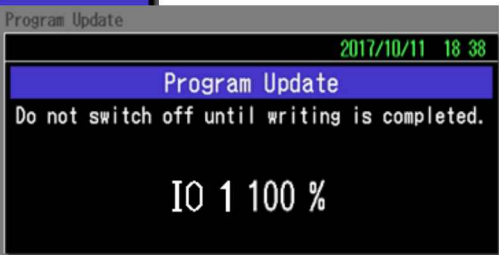
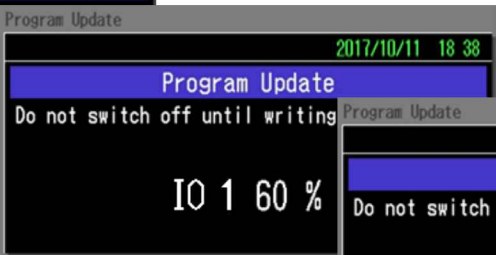
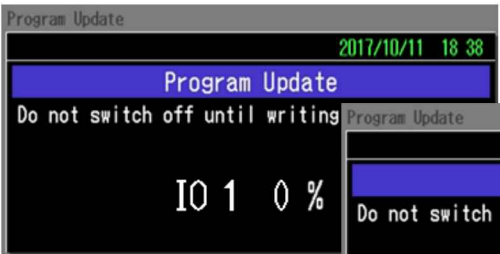


Contact to ISHIDA agent for the password required in the level 3. "???" is displayed on the password displayed screen for security reasons.



5. Press  key.

- The program update (installation) starts.



6. A message "Completed. Turn off the power and restart." appears. As indicated, turn off the power switch.



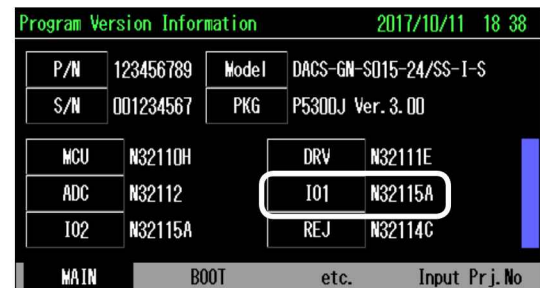
7. Return the switches DIP-SW1-8 on the board to the OFF state.

8. Turn ON the power switch again.

- Point the cursor to [Setup Menu] - [Program Version Information], and press key.



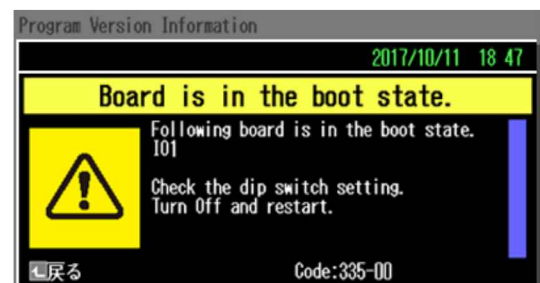
9. Check that the corresponding program No. is displayed on the Program Version Information screen.



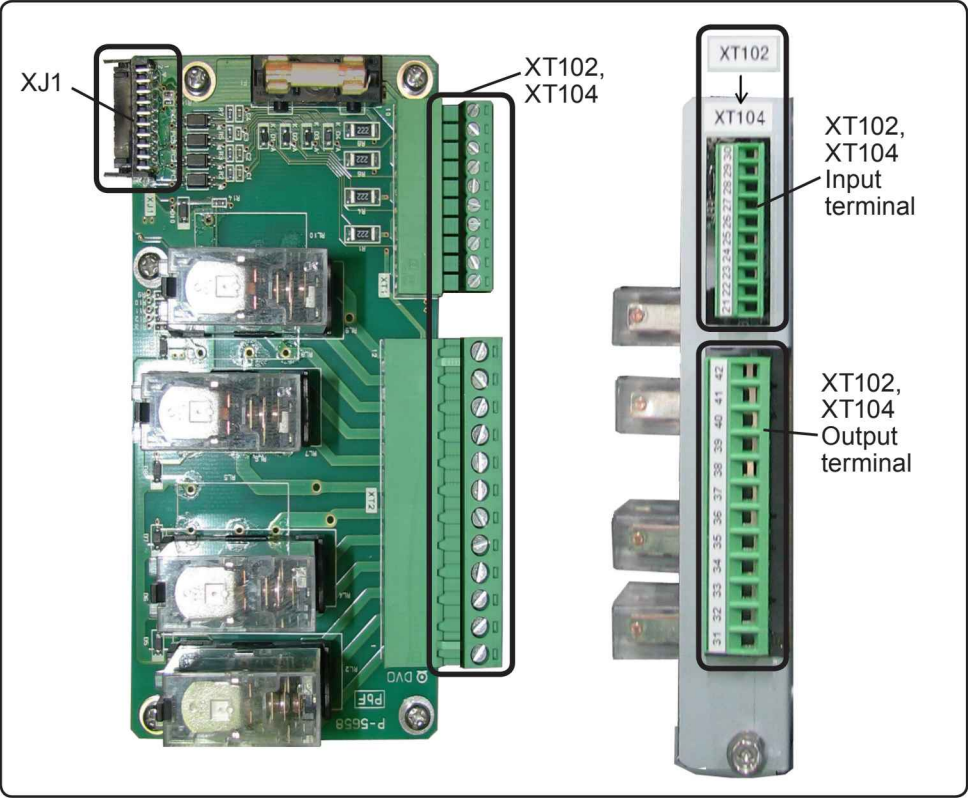
■ Function to detect the boot state of the board program [additional function]

This function detects that the board program has started in the boot state (in which the program update is ready). In the boot state, normal operation is not available. There are two causes for the boot operation of the board program:

- After downloading the program, power is turned ON soon after OFF.
(The residual voltage of the switching power supply disables the power OFF.)
- The switches DIP-SW1 to 8 on the board are remained to be ON.



4.8.1 Relay board P-5658* [XT102, XT104]



■ Board Functions

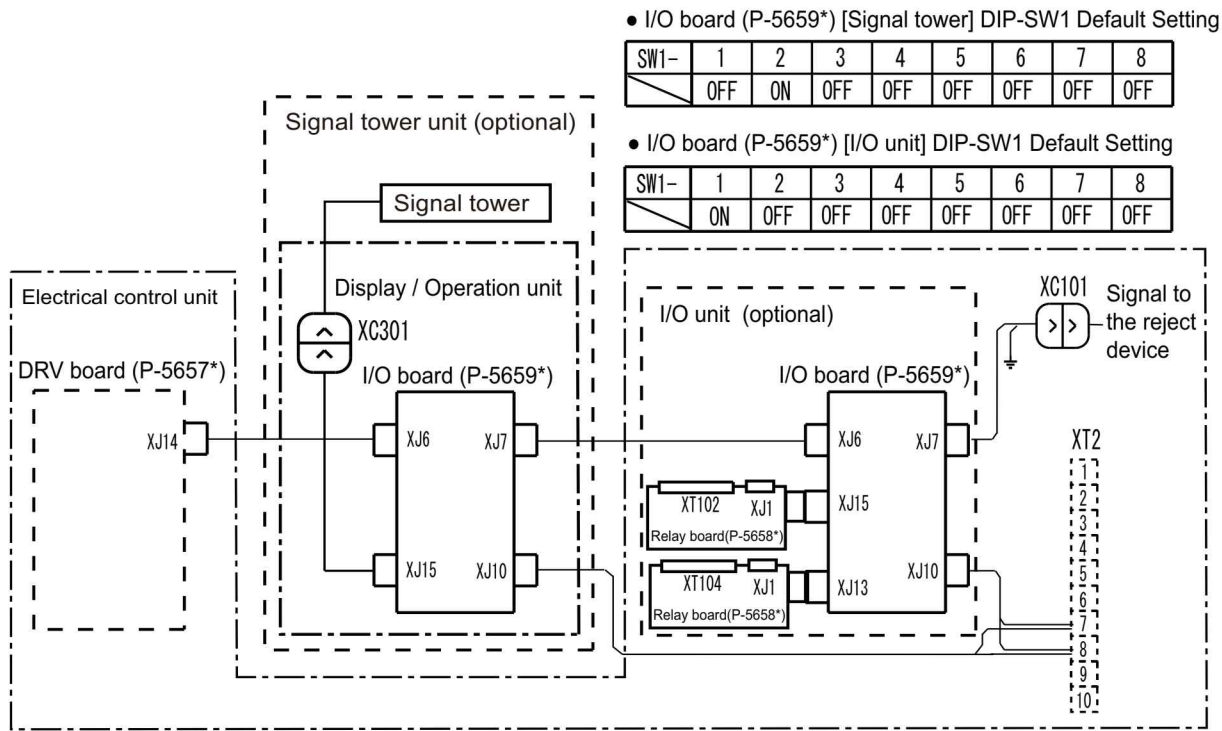
- Transmits external output signal
- Receives external input signal
- Communication with I/O board (P-5659*)

■ Connector usage classification

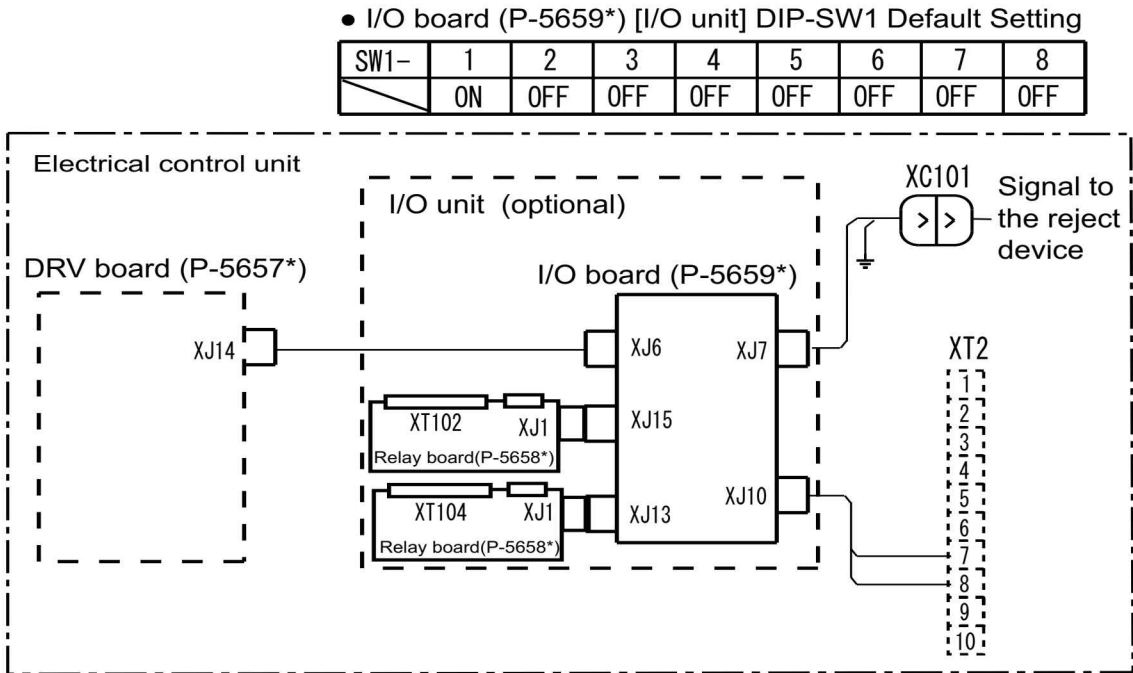
CONNECTOR No.		FUNCTION
XJ1	20P	Communication with DRV board (P-5657*)
XT102, XT104	10P	Input terminal
XT102, XT104	12P	Output terminal

4.8.2 Realy terminal unit (optional)

■ With signal tower



■ Without signal tower



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